

1.2 Physical Design

1.2.1 Dimension Tables

BOOK_DIM

```
CREATE TABLE book_dim
(
    book_key          NUMBER          NOT NULL,
    bookID            VARCHAR2(6)     NOT NULL,
    copyID            VARCHAR2(7)     NOT NULL,
    bookTitle          VARCHAR2(200)  NOT NULL,
    bookImei           VARCHAR2(50),
    language           VARCHAR2(50)   NOT NULL,
    author             VARCHAR2(100)  NOT NULL,
    publishYear        NUMBER(4)      NOT NULL,
    bookCategory       VARCHAR2(50)   NOT NULL,
    publisherName       VARCHAR2(100) NOT NULL,
    publisherLocation   VARCHAR2(100) NOT NULL,
    bookStatus         VARCHAR2(20)   NOT NULL,
    bookCondition       VARCHAR2(20)   NOT NULL,
    CONSTRAINT book_dim_pk PRIMARY KEY(book_key),
    CONSTRAINT book_dim_bookid_fk FOREIGN KEY (bookID) REFERENCES BOOK (BOOKID)
);
```

MEMBER_DIM

```
CREATE TABLE member_dim
(
    member_key          NUMBER          NOT NULL,
    memberID            VARCHAR2(6)     NOT NULL,
    memberName          VARCHAR2(100)   NOT NULL,
    memberEmail         VARCHAR2(100)   NOT NULL,
    dob                 DATE             NOT NULL,
    contactNum          VARCHAR2(20)    NOT NULL,
    memberCountry       VARCHAR2(50)    NOT NULL,
    memberState         VARCHAR2(50)    NOT NULL,
    memberCity          VARCHAR2(50)    NOT NULL,
    memberZipcode       VARCHAR2(10)    NOT NULL,
    effective_start_date DATE           NOT NULL,
    effective_end_date   DATE           NOT NULL,
    is_current_flag      CHAR(1)        NOT NULL,
    CONSTRAINT member_dim_pk PRIMARY KEY (member_key),
    CONSTRAINT member_dim_memberid_fk FOREIGN KEY (memberID) REFERENCES MEMBER
(MEMBERID)
);
```

DATE_DIM

```

CREATE TABLE DATE_DIM
(
    DATE_KEY          NUMBER          NOT NULL,
    CAL_DATE          DATE            NOT NULL,
    FULL_DESC         VARCHAR2(50),
    DAY_WEEK          VARCHAR2(10),
    DAY_NUM_MONTH     NUMBER(2),
    DAY_NUM_YEAR      NUMBER(3),
    LAST_DAY_IND      CHAR(1),
    CAL_WEEK_END_DATE DATE,
    CAL_WEEK_YEAR     NUMBER(2),
    CAL_MONTH_NAME    VARCHAR2(15),
    CAL_MONTH_YEAR    NUMBER(2),
    CAL_YEAR_MONTH    CHAR(7),
    CAL_QUARTER       CHAR(2),
    CAL_YEAR_QUARTER  CHAR(7),
    CAL_YEAR          NUMBER(4),
    HOLIDAY_IND       CHAR(1),
    WEEKDAY_IND       CHAR(1),
    EXAM_WEEK_IND     CHAR(1),
    STUDY_WEEK_IND    CHAR(1),
    FESTIVE_EVENT     VARCHAR2(50), -- Added to store holiday names
    CONSTRAINT DATE_DIM_PK PRIMARY KEY (DATE_KEY)
);

```

STAFF_DIM

```

CREATE TABLE staff_dim
(
    staff_key          NUMBER          NOT NULL,
    staffID            VARCHAR2(5)     NOT NULL,
    staffName          VARCHAR2(100)   NOT NULL,
    staffEmail         VARCHAR2(100)   NOT NULL,
    staffPhoneNum      VARCHAR2(20)    NOT NULL,
    department         VARCHAR2(50)    NOT NULL,
    position           VARCHAR2(50)    NOT NULL,
    hireDate           DATE            NOT NULL,
    effective_start_date DATE          NOT NULL,
    effective_end_date DATE          NOT NULL,
    is_current_flag    CHAR(1)         NOT NULL,
    CONSTRAINT staff_dim_pk PRIMARY KEY (staff_key),
    CONSTRAINT staff_dim_staffid_fk FOREIGN KEY (staffID) REFERENCES STAFF
(STAFFID)
);

```

SUPPLIER_DIM

```

CREATE TABLE supplier_dim
(
    supplier_key       NUMBER          NOT NULL,

```

```

supplierID          VARCHAR2(5)    NOT NULL,
supplierName        VARCHAR2(100)  NOT NULL,
supplierLocation    VARCHAR2(100)  NOT NULL,
supplierContactNum  VARCHAR2(20)   NOT NULL,
supplierEmail       VARCHAR2(100)  NOT NULL,
effective_start_date DATE          NOT NULL,
effective_end_date  DATE          NOT NULL,
is_current_flag     CHAR(1)        NOT NULL,
CONSTRAINT supplier_dim_pk PRIMARY KEY (supplier_key),
CONSTRAINT supplier_dim_supplierid_fk FOREIGN KEY (supplierID) REFERENCES
SUPPLIER (SUPPLIERID)
);

```

1.2.1 Fact Tables

RESERVATION_FACT

```

CREATE TABLE RESERVATION_FACT
(
    member_key          NUMBER NOT NULL,
    book_key            NUMBER NOT NULL,
    staff_key           NUMBER NOT NULL,
    reserve_start_date_key NUMBER NOT NULL,
    reserve_end_date_key NUMBER NOT NULL,
    reserveID           VARCHAR2(7) NOT NULL,
    reservationStatus   VARCHAR2(20) NOT NULL,
    reservationDuration NUMBER,      -- calculated days between start and end
    CONSTRAINT RF_MEMBER_FK FOREIGN KEY(member_key) REFERENCES
MEMBER_DIM(member_key),
    CONSTRAINT RF_BOOK_FK   FOREIGN KEY(book_key)   REFERENCES
BOOK_DIM(book_key),
    CONSTRAINT RF_STAFF_FK  FOREIGN KEY(staff_key)  REFERENCES
STAFF_DIM(staff_key),
    CONSTRAINT RF_START_DATE_FK FOREIGN KEY(reserve_start_date_key)
REFERENCES DATE_DIM(date_key),
    CONSTRAINT RF_END_DATE_FK FOREIGN KEY(reserve_end_date_key)
REFERENCES DATE_DIM(date_key),
    CONSTRAINT RF_RESERVEID_FK FOREIGN KEY(reserveID) REFERENCES
Reservation(reserveID),
    CONSTRAINT RESERVATION_FACT_PK PRIMARY KEY(member_key, book_key, staff_key,
reserve_start_date_key, reserveID)
);

```

LOAN_FACT

```

CREATE TABLE LOAN_FACT
(
    member_key          NUMBER NOT NULL,
    book_key            NUMBER NOT NULL,

```

```

    staff_key          NUMBER NOT NULL,
    date_key           NUMBER NOT NULL,
    loanID             VARCHAR2(8) NOT NULL,
    loanDuration       NUMBER NOT NULL,
    overdueDays        NUMBER,
    loanStatus         VARCHAR2(20) NOT NULL,
    totalFine          NUMBER(8,2),
    fine_paid_flag     NUMBER(1),
    CONSTRAINT LF_MEMBER_FK FOREIGN KEY(member_key) REFERENCES
MEMBER_DIM(member_key),
    CONSTRAINT LF_BOOK_FK  FOREIGN KEY(book_key)   REFERENCES
BOOK_DIM(book_key),
    CONSTRAINT LF_STAFF_FK FOREIGN KEY(staff_key)  REFERENCES
STAFF_DIM(staff_key),
    CONSTRAINT LF_DATE_FK  FOREIGN KEY(date_key)   REFERENCES
DATE_DIM(date_key),
    CONSTRAINT LF_LOANID_FK FOREIGN KEY(loanID)     REFERENCES Loan(loanID),
    CONSTRAINT LOAN_FACT_PK PRIMARY KEY(member_key, book_key, staff_key,
date_key, loanID)
);

```

PURCHASE_FACT

```

CREATE TABLE PURCHASE_FACT
(
    date_key          NUMBER NOT NULL,
    book_key          NUMBER NOT NULL,
    staff_key         NUMBER NOT NULL,
    supplier_key      NUMBER NOT NULL,
    purchaseID        VARCHAR2(10) NOT NULL,
    purchaseQuantity  NUMBER NOT NULL,
    purchaseUnitCost  NUMBER(10,2) NOT NULL,
    purchaseTotalCost NUMBER(12,2) NOT NULL,
    CONSTRAINT PF_DATE_FK  FOREIGN KEY(date_key)   REFERENCES
DATE_DIM(date_key),
    CONSTRAINT PF_BOOK_FK  FOREIGN KEY(book_key)   REFERENCES
BOOK_DIM(book_key),
    CONSTRAINT PF_STAFF_FK FOREIGN KEY(staff_key)  REFERENCES
STAFF_DIM(staff_key),
    CONSTRAINT PF_SUPPLIER_FK FOREIGN KEY(supplier_key) REFERENCES
SUPPLIER_DIM(supplier_key),
    CONSTRAINT PF_PURCHASEID_FK FOREIGN KEY(purchaseID) REFERENCES
PURCHASE(purchaseID),
    CONSTRAINT PURCHASE_FACT_PK PRIMARY KEY(date_key, book_key, staff_key,
supplier_key, purchaseID)
);

```

SALES_FACT

```

CREATE TABLE SALES_FACT

```

```
(
    date_key          NUMBER          NOT NULL,
    member_key        NUMBER          NOT NULL,
    book_key          NUMBER          NOT NULL,
    staff_key         NUMBER          NOT NULL,
    orderID           VARCHAR2(7)    NOT NULL,
    orderQty          NUMBER          NOT NULL,
    orderUnitPrice    NUMBER(10,2)    NOT NULL,
    orderTotalPrice   NUMBER(12,2)    NOT NULL,
    CONSTRAINT SF_DATE_FK    FOREIGN KEY(date_key)    REFERENCES
DATE_DIM(date_key),
    CONSTRAINT SF_MEMBER_FK FOREIGN KEY(member_key)   REFERENCES
MEMBER_DIM(member_key),
    CONSTRAINT SF_BOOK_FK   FOREIGN KEY(book_key)    REFERENCES
BOOK_DIM(book_key),
    CONSTRAINT SF_STAFF_FK  FOREIGN KEY(staff_key)   REFERENCES
STAFF_DIM(staff_key),
    CONSTRAINT SF_ORDERID_FK FOREIGN KEY(orderID)    REFERENCES ORDERS(orderID),
    CONSTRAINT SALES_FACT_PK PRIMARY KEY(date_key, member_key, book_key,
staff_key, orderID)
);
```